

Design Memory or total capacity 64MB with word size 32 bits
Using RAM Chips Size 16MB x 16 Bit

total capacity = 64MB word size = 32 bits
chip size = 16MB x 16 Bit

① word size = $\frac{32}{8} = 4$ Bytes

word = $\frac{\text{total capacity}}{\text{size word}} = \frac{64\text{MB}}{4\text{B}} = 16\text{MB}$

Target

Target = 16MB x 32

Row = $\frac{16\text{MB}}{16\text{MB}} = 1$ Columns = $\frac{32}{16} = 2$

number of RAM chip = $2 \times 1 = 2$

address line for each RAM chip = $2^4 \times 2^{20} = 2^{24}$
 $A_0 \rightarrow A_{23}$

" " for all memory = $2^4 \times 2^{20} = 2^{24}$
 $A_0 \rightarrow A_{23}$

number of data line for all memory = 32 $D_0 \rightarrow D_{31}$

" " " " " each chip = 16 $D_0 \rightarrow D_{15}$

number of selector = $24 - 24 = 0$

No Decoder is need only one Row

0×1

Design memory or total Capacity 128 MB with word size 16 bits using RAM chips size 32 M x 8

$$\textcircled{1} \text{ Size word} = \frac{16}{8} = 2 \text{ Byte}$$

$$\text{Target words} = \frac{128 \text{ MB}}{2 \text{ B}} = 64 \text{ MB words}$$

$$\text{Target Capacity} = 64 \text{ M} \times 16$$

$$\text{Row} = \frac{64 \text{ M}}{32 \text{ M}} = 2 \quad \text{Columns} = \frac{16}{8} = 2$$

$$\text{number of RAM chips} = 2 \times 2 = 4 \text{ chips}$$

$$\begin{array}{ccccccc} \text{Number of Address line for each chip} & = & 2^5 \times 2^{20} & = & 2^{25} & A_0 \rightarrow A_{24} \\ \text{" " " " " " " " For all memory} & = & 2^6 \times 2^{20} & = & 2^{26} & A_0 \rightarrow A_{25} \end{array}$$

$$\text{Decoder} = 2^6 - 2^5 = 1$$

$$1 \times 2^1 = 1 \times 2$$

$$\text{Data line of All memory} = 16 \quad \begin{array}{c} D \\ A_0 \rightarrow A_{15} \end{array}$$

$$\text{" " " each chip} = 8 \quad \begin{array}{c} D \\ A_0 \rightarrow A_7 \end{array}$$

طریقی سے جانو

اری والی بتو فیکر ولکم بالمش

Design Rom memory total capacity 64 Kilo words
word size 32 bits Using chips size 16K X 16

total capacity 64 K words

Target 64 K * 32
chip size 16 K X 16

$$\text{Row} = \frac{64K}{16} = 4$$

$$\text{Columns} = \frac{32}{16} = 2$$

$$\text{Number of Chips} = 4 \times 2 = 8$$

$$\text{Number of Address line of all memory} = 64 \times 2^{10} = 2^6 \times 2^{10} = 2^{16}$$

$$A_0 \rightarrow A_{15}$$

$$\text{Number of Address line of each chip} = 16 \times 2^{10} = 2^4 \times 2^{10} = 2^{14}$$

$$A_0 \rightarrow A_{13}$$

$$\text{Decoder} = 16 - 14 = 2$$

$$2 \times 2^2 = 2 \times 4$$

What are the control signal required for this memory? Read only



BE

Design memory system with $4M \times 8 \text{ bit}$
Capacity $256K \times 8 \text{ bit}$ RAM chips

$$\text{Target memory} = 4 \times 2^{20} = 2^{22} \times 8 \text{ bit}$$

$$\text{USmall " } = 2^8 \times 2^{10} = 2^{18} \times 8 \text{ bit}$$

$$\text{Row} = \frac{2^{22}}{2^{18}} = 2^4 = 16 \quad \text{Columns} = \frac{8}{8} = 1$$

$$\text{number of chips} = 16 \times 1 = 16 \text{ chips}$$

$$\text{Data lines for Target memory} = 22 \quad A_0 \rightarrow A_{21}$$

$$\text{" " " chip} = 18 \quad A_0 \rightarrow A_{17}$$

$$\text{Decoder} = 22 - 18 = 4$$

$$4 \times 2^4 = 4 \times 16$$

طرح من سیدنا محمد

افتمکروا اب وجبتی خدایکم

ادعوا لی بتوفیق ولکم بالملک